MILITARY SPECIFICATION

PROJECTILE, 8 INCH, M424A1 LOADING ASSEMBLY

1. SCOPE

1.1 This specification covers requirements, quality assurance provisons, and the preparation for delivery criteria for one type of Projectile, 8 Inch, M424A1 Loading Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATION

FEDERAL

C-F-206 - Felt, Sheet, Wool, Pressed

MILITARY

MIL-B-2427 - Box Ammunition Packing, Wood Nailed
MIL-C-401 - Composition B
MIL-P-50560(MU) - Projectile, HES, 8 Inch, XM424E1 Metal Parts
MIL-R-11470 - Radiographic Inspection: Qualification Equipment, Operations and Procedures

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1167 - Ammunition Data Card
MIL-STD-1168 - Lot Numbering of Ammunition

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DRAWINGS

ORDNANCE CORPS

8787063 - Box Packing Ammo, for Projectile 8 Inch M424 or M424A1
8787847 - Washer
8787848 - Cup, Fuze Well
8795639 - Gasket
9241673 - Projectile, 8 Inch, M424A1 Loading Assembly
ET-9241673 - Equipment Tabulation

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The projectile shall comply with all requirements specified on the drawing (dwg) 9241673, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 X-Ray examination. Compliance with these requirements shall be determined in accordance with 4.4.1.

3.4.1 Cracks. There shall be no cracks (cooling fissures) in the powder charge, over 1/32 inch wide, as determined by visual examination of the x-rays and the visable surface of the powder charge as poured in the projectile after cooling. In addition, there shall be no more than 3 cracks total of 1/32 inch in maximum width in the complete charge. Only 2 cracks will be permitted in the bottom segment of the charge.

3.4.1.1 Complete separation of charge. There shall be no evidence of complete separation of the powder charge regardless of crack width.

3.4.2 Cavities. No individual cavity, excluding pipes, porous areas, cracks, and annular rings, shall exceed a projected dimension of 3/4 inch. Cavitations not exceeding

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1/2 inch width will be acceptable up to one inch in length. Total summation of all cavitation shall not exceed one square inch. Cavities in the base of the fuze well shall not be cause for rejection provided the sum of the areas of the cavities is not greater than 80 percent of the area of the fuze well base and no individual point extends more than 1/4 inch below the maximum depth of the fuze well as permitted by the drawing. Low breaks in the bottom of the fuze well cavity caused by the removal of the former funnel, shall be considered as a cavity when determining the sum of the areas and the maximum depth as permitted by the above criterion. In addition, low breaks (located at the top surface of the charge, excluding the fuze well bottom surface) caused by the removal of the former funnels, shall not exceed 1/4 inch in depth when visibly examined.

3.4.3 Pipes and porous areas. Pipes and porous areas shall be treated as cavities, and shall not be subjected to restrictions placed upon cavities (see 3.4.2), except that 80 percent of the projected length and 80 percent of the projected area shall be considered for acceptance purposes.

3.5 Rotating band damage. The rotating band shall be examined for the following defects (see 4.4.3):

a. Cracks
b. Foreign matter embedded in rotating band material
c. Any dent, nick, scratch or gouge which exceeds the allowable depth, see Fig. I.

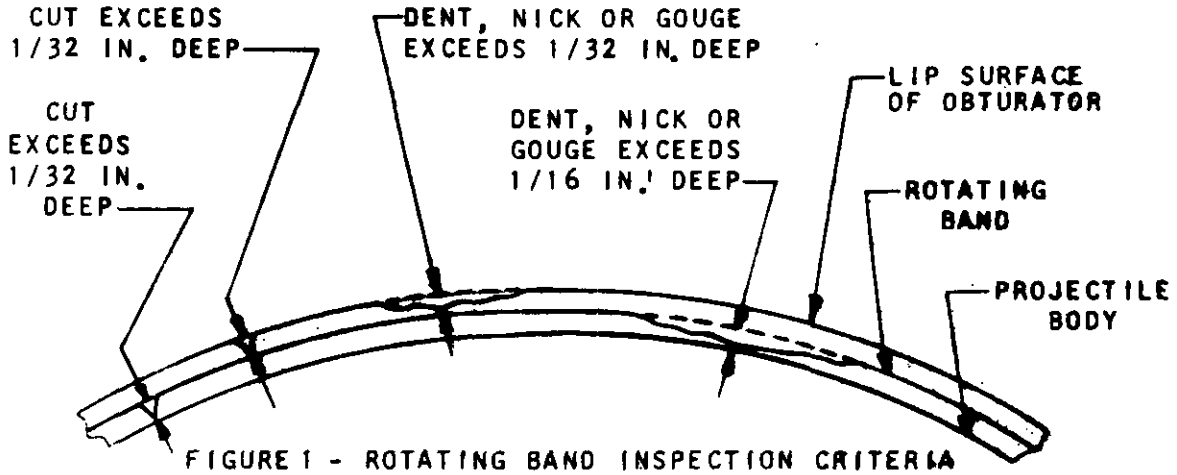


FIGURE 1 - ROTATING BAND INSPECTION CRITERIA

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3.6 Workmanship.

3.6.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.6.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.6.3 Burrs. Parts shall be free from burrs.

3.6.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.6.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. (see 6.4) When the contractor requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg 9241673) and 3 sets of parts for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests need not be conducted in order specified in Table I.

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Table I
First Article Inspections and Tests

Inspection or Test	Contractor's Inspection Permissable Before Submission	Non Dest	Parts	Assy
1. Classification of Characteristics	X	X	X	X
2. X-ray Examination (see 3.4)	X	X	X	X
3. Chemical composition of powder test (see 4.3.3.3)	X	X	X	
4. Thickness of washer (see 4.3.3.2)	X	X	X	

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Product acceptance inspection.

4.3.1 Lot formation. Lot formation shall be in accordance with MIL-STD-1168. Changes in lot numbers shall be recorded on Ammunition Data Cards in accordance with MIL-STD-1167 and MIL-STD-1168.

4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Washer (see dwg 8787847) covering a detail of dwg 9241673.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Thickness (see 4.3.3.2)	Gage
	102. Inside diameter, max	Gage
	103. Outside diameter, min	Gage
	104. Material (see 3.1)	Visual
	105. No splits, cracks or holes	Visual
Minor:	None defined.	

4.3.2.1.2 Cup, fuze well (see dwg 8787848) covering a detail of dwg 9241673.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Outside diameter of cup, min	Gage
	102. True position of outside diameter of cup	Gage
	103. Thickness through bottom	Gage
	104. Inside depth	Gage

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Categories	Characteristics	Method of Inspection
Major: (Continued)		
	105. Inside diameter of cup	Gage
	106. Pitch diameter of thread, min	Gage
	107. Major diameter of thread, min	Gage
	108. Material (see 3.1)	Visual
	109. Objective evidence (see 6.2 for protective finish in accordance with dwg 8787848	Visual
Minor:		
	201. Finish	Visual
	202. Workmanship (see 3.6)	Visual/Manual

4.3.2.1.3 Gasket (see dwg 8795639) covering a detail of dwg 9241673.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. No tear, split, or hole	Visual
	102. Material (see 3.1)	Visual
Minor:		
	201. Workmanship (see 3.6)	Visual/Manual

4.3.2.1.4 Projectile, 8 Inch, M42A1 Loading Assembly (see dwg 9241673).

Categories	Characteristics	Method of Inspection
Critical:		
	1. Any evidence of powder chips in fuze well cavity	Visual
	2. Weight of projectile less grommet	Gage
Major:		
	101. Objective evidence (see 6.2) for metal parts assembly in accordance with MIL-P-50560(MU)	Visual
	102. Location of center of gravity marking	Gage
	103. Diameter of charge cavity	Gage
	104. True position of diameter of charge cavity	Gage
	105. Depth of charge cavity	Gage

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Categories	Characteristics	Method of Inspection
Major: (continued)		
106.	Depth of washer cavity	Gage
107.	Bourrelet diameters after painting and marking	Gage
108.	Depth from plate counterbore bottom to inside of fuze well cup, min	Gage
109.	Marking	Visual
110.	Sealing compound applied to threads specified	Visual
111.	Painting (no bare spots)	Visual
112.	Dents or nicks on nose plug exceeding 1/16 inch depth	Gage
113.	Dents or nicks on ogive exceeding 1/16 inch depth	Gage
114.	Presence of rubber gasket between ogive and housing	Visual
115.	Ogive assembled hand tight	Manual
116.	No damage to rotating band (see 3.5)	Visual/Gage
117.	Grommet properly assembled	Visual/Manual
118.	Materials (see 3.1)	Visual
Minor:		
201.	Workmanship (see 3.6)	Visual/Manual

4.3.2.1.5 Box, packing, ammunition for Projectile, 8 Inch, M424 or M424A1 (see dwg 8787063) before closing cover.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	No pads missing	Visual
102.	No supports missing	Visual
103.	Sling properly assembled not damaged	Visual/Manual
104.	Projectile tight in box	Manual
105.	Fuze container tight and padded properly	Manual
106.	Supports not split or cracked	Visual
107.	Pads cemented to wood	Manual
108.	Location of center spacer	Measure*
109.	Nose support assembly flush with end of box	Visual/Manual
110.	Materials (see 3.1)	Visual

*Measure with suitable scale.

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Categories	Characteristics	Method of Inspection

Minor: None defined.

4.3.2.1.6 Box, packing, ammunition, for Projectile, 8 Inch, M424 or M424A1 (see dwg 8787063).

Categories	Characteristics	Method of Inspection

Critical:
1. Car seal present and properly attached...Visual

Major:
101. No hardware missing, broken or loose.Visual/Manual
102. No strapping missing, broken or loose.....Visual/Manual
103. No boards broken or split in accordance with MIL-B-2427.....Visual
104. Marking.....Visual
105. Materials (see 3.1).....Visual
106. No contents loose in box.....Manual

Minor:
201. Workmanship (see 3.6).....Visual/Manual

4.3.3 Testing. All tests shall be performed. Tests need not be performed in the order specified.

Test	Test Classification
a. X-ray examination	Major
b. Thickness of washer	Major
c. Chemical composition of powder	Major

Of these tests c is considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 X-ray examination (see 3.4). After each shut down of 5 days or more of the loading line, a sample of 3 projectiles shall be selected at random from the first days production for x-ray and visual examination. A sample of 3 projectiles for x-ray shall be also selected from the first days production of each new powder run. Any housing that does not comply with 3.4 shall be rejected and removed from the lot and rejected. Any defective lot shall be subjected to 100 percent x-ray and visual examination to determine if any additional defective housings are present. Upon resuming the pouring operation an additional sample consisting of the first 6 housings shall be submitted to x-ray and visual examination and shall comply with 3.4, before continuing regular production operations (see 4.4.1).

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4.3.3.2 Thickness of washer (see dwg 8787847). A sample shall be selected at random in accordance with Table IV. The sample shall be tested in accordance with 4.4.2. If any item fails to comply with the drawing, the lot shall be rejected.

Table IV

Sample sizes for destructive testing

Number Required to be Delivered	Sample Size for No Failures
1-10	15
11-18	16
19-25	17
26-37	18
38-57	19
58-105	20
106-249	21
250 and over	22

Lot Size Required = Number Required to be Delivered + Sample Size.

4.3.3.3 Chemical composition of powder. The Government inspector shall designate selected samples for test in accordance with Testing, MIL-C-401. If any sample fails to comply with Testing, MIL-C-401, the lot shall be rejected.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). ELs for assemblies are listed on the Equipment Tabulation, ET-9241673. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired. (See 6.3)
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL. (See 6.1.2)

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4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accurancy tolerances of the test equipment used.

4.4.1 X-ray examination (see 3.4). The sample loaded housing shall be x-rayed with approved x-ray equipment in accordance with MIL-R-11470. The sample projectiles shall be x-rayed on 2 planes 90 degrees apart. View showing largest dimention of cavitation shall be used for computation of characteristics. However, if the original x-ray does not indicate any imperfection, as determined by the Government inspector, an x-ray in the second plane will not be required.

4.4.2 Thickness of washer (see 4.3.3.2). The thickness of the washer shall be measured with equipment specified in C-F-206. Ten measurements shall be averaged and the resultant average used for acceptance.

4.4.3 Rotating band damage (see 3.5). An impression with "cast stone" or "grey stone" shall be taken of each nick, scratch, dent, or gouge to determine the depth of the imperfection by an optical comparator type device or any accurate measuring device approved by Picatinny Arsenal, Dover, New Jersey 07801, Attention: SMUPA-QA-N-I. For allowable depth of the imperfection, see Fig I. In addition, the rotating band shall be visually inspected for any crack or foreign matter embedded in the rotating band.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The projectile shall be packaged in accordance with dwg 8787063 using the specified moisture barrier.

5.1.2 Level C. The projectile shall be packaged in accordance with dwg 8787063 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The projectile shall be packed in accordance with dwg 8787063.

5.3 Marking. Marking shall be as specified in accordance with dwg 8787063.

5.4 Data cards. Data cards shall be prepared in accordance with MIL-STD-1167.

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6. NOTES

6.1 Ordering data. Procurement documents should specify:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1 and 5.4). Data specified in 4.3.4.1 and 5.4 will be delivered as identified on a numbered DD Form 1667 when specified on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-QA-N-I.

6.4 Instructions for first article inspection (see 4.2.1). Parts may be used in subassemblies if the examination or tests (see 4.2.2, Table I) are verified by a representative designated by the Government. Subassemblies may be used in final assemblies if the examination or tests (see 4.2.2, Table I) are verified by a representative designated by the Government.

6.5 Supersession data. This specification includes the requirements of Purchase Description PA-PD-924 (Rev 1).

Custodian:
Army - MU

Preparing activity:
Army - MU

Project No. 1320-A119

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. *Fold on lines on reverse side, staple in corner, and send to preparing activity.* Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-P-50562(MU) Projectile, 8 inch, M424A1 Loading Assembly

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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